

Instruction Set					
Data Transfer					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x00	LDA	Memory address	12	Load accumulator from memory	A = *OP;
0x01	LDI	Immediate data	8	Load accumulator from immediate	A = OP;
0x02	STA	Memory address	12	Store accumulator to memory	*OP = A;
0x03	LIA	None	9	Load indirect access	A = *A;
Logic					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x04	NOT	None	5	Bitwise NOT	A = ~A;
0x05	AND	Memory address	13	Bitwise AND	A = A & *OP;
0x06	OR	Memory address	13	Bitwise OR	A = A *OP;
0x07	XOR	Memory address	13	Bitwise XOR	A = A ^ *OP;
0x08	SHL	None	5	Logic left shift	A = A << 1;
0x09	SHR	None	5	Logic right shift	A = A >> 1;
Arithmetic					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x0A	ADD	Memory address	13	Add memory to accumulator	A = A + *OP;
0x0B	SUB	Memory address	13	Subtract memory from accumulator	A = A - *OP;
0x0C	CMP	Memory address	13	Compare accumulator and memory	EQUAL = (A == *OP) ? 1 : 0; COUT = (A <= *OP) ? 1 : 0;
Jump					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x0D	JMP	Memory address	8	Unconditional jump	PC = OP;
0x0E	JPE	Memory address	8 (5)	Jump if accumulator is equal memory	PC = (EQUAL == 1) ? OP : PC;
0x0F	JPA	Memory address	8 (5)	Jump if accumulator is above memory	PC = (EQUAL == 0 && COUT == 0) ? OP : PC;
0x10	JPB	Memory address	8 (5)	Jump if accumulator is below memory	PC = (EQUAL == 0 && COUT == 1) ? OP : PC;
0x11	CALL	Memory address	13	Jump to subroutine	*SP = PC; SP--; PC = OP;
0x12	RET	None	9	Return from subroutine	SP++; PC = *SP;
Stack					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x13	LSP	None	5	Load stack pointer from accumulator	SP = A;
0x14	PUSH	None	8	Push accumulator to stack	*SP = A; SP--;
0x15	POP	None	9	Pop stack to accumulator	SP++; A = *SP;
I/O					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x16	IN	I/O address	12	Read I/O data	A = *OP; /* (M/IO = 0) */
0x17	OUT	I/O address	12	Write I/O data	*OP = A; /* (M/IO = 0) */
Idle					
Opcode	Mnemonic	Operand	Cycles	Description	Algorithm (C syntax)
0x18	NOP	None	4	No operation	;
0x19	HALT	None	4 (∞)	Halt CPU	while(1);

ALU Inputs

Operation (C syntax)	Description	Inputs							
		MODE	CIN	S3	S2	S1	S0	SHR	OE
~A	Bitwise complement	1	X	0	0	0	0	1	0
A & B	Bitwise logic AND	1	X	1	0	1	1	1	0
A B	Bitwise logic OR	1	X	1	1	1	0	1	0
A ^ B	Bitwise logic XOR	1	X	0	1	1	0	1	0
A + B	Arithmetic addition	0	1	1	0	0	1	1	0
A - B	Arithmetic subtraction	0	0	0	1	1	0	1	0
A << 1	Left logic bit shift	0	1	1	1	0	0	1	0
A >> 1	Right logic bit shift	X	X	X	X	X	X	0	1
A - B - 1	Compare operands	0	1	0	1	1	0	1	0

ALU Outputs

Comparison Result	Outputs	
	ALU_COUT	ALU_EQUAL
A > B	0	0
A = B	1	1
A < B	1	0

Control Inputs

Bit Index	Signal	Active Level	Description
Arithmetic Logic Unit			
0	COUT	L	ALU's carry output (COUT output)
1	EQUAL	H	ALU's equal operands (EQUAL output)
Control Unit			
2-5	IC[0..3]	-	CU's instruction counter
Instruction Register			
6-10	IR[0..4]	-	IR's data (IREG[0..4] output)

Control Outputs

Bit Index	Signal	Active Level	Description
Accumulator			
0	AI	L	Accumulator's data input enable (IE input)
1	AO	L	Accumulator's data output enable (OE input)
B Register			
2	BI	L	B Register's data input enable (IE input)
Arithmetic Logic Unit			
3	RO	L	ALU's result output enable (OE input)
4	SHR	L	ALU's right logic shift output enable (SHR input)
5	S0	H	ALU's operation selection 0 (S0 input)
6	S1	H	ALU's operation selection 1 (S1 input)
7	S2	H	ALU's operation selection 2 (S2 input)
8	S3	H	ALU's operation selection 3 (S3 input)
9	CIN	L	ALU's carry input (CIN input)
10	MODE	H	ALU's operation mode (MODE input)
Instruction Register			
11	IRI	L	IR's data input enable (IE input)
Program Counter			
12	PCI	L	PC's data input enable (IE input)
13	PCDO	L	PC's data output enable (DOE input)
14	PCAO	L	PC's address output enable (AOE input)
15	PCCE	H	PC's count enable (CE input)
Stack Pointer			
16	SPI	L	SP's data input enable (IE input)
17	SPO	L	SP's data output enable (OE input)
18	SPCE	L	SP's count enable (CE input)
19	SPU/SPD	H	SP's counter direction (UP/DOWN input)
Data Address Register			
20	DARI	L	DAR's data input enable (IE input)
21	DARO	L	DAR's address output enable (OE input)
Bus Interface Unit			
22	DI/DO	H	BIU's data pins direction (DI/DO input)
23	DEN	L	BIU's data pins enable (DEN input)
24	ACO	L	BIU's address and control pins output enable (ACO input)
25	ALE	L	BIU's address pins latch enable (ALE input)
26	M/IO	L	BIU's memory/IO operation pin (M/IO_BUS input)
27	RD	L	BIU's read operation pin (RD_BUS input)
28	WR	L	BIU's write operation pin (WR_BUS input)
Control Unit			
29	ILC	L	CU's instruction last cycle (ILC signal)

Inputs				Active Outputs
IR[0..4]	IC[0..3]	EQUAL	COUT	
Reset				
X	0	X	X	
Read Cycle				
X	n	X	X	[M/IO], ACO, ALE, PCAO DARO SPO
X	n+1	X	X	[M/IO], ACO, DEN, RD
X	n+2	X	X	[M/IO], ACO, DEN, RD, AI BI IRI DARI PCI SPI
X	n+3	X	X	[M/IO], ACO, [PCCE], [ILC]
Write Cycle				
X	n	X	X	[M/IO], ACO, DI/DO, ALE, PCAO DARO SPO
X	n+1	X	X	[M/IO], ACO, DI/DO, DEN, WR, PCDO AO
X	n+2	X	X	[M/IO], ACO, DI/DO, DEN, PCDO AO
X	n+3	X	X	[M/IO], ACO, [SPCE], [SPU/SPD], [ILC]
Opcode Fetch				
X	1..4	X	X	Read(PCAO, IRI, PCCE)
Execution LDA				
0x00	5..8	X	X	Read(PCAO, DARI, PCCE)
0x00	9..12	X	X	Read(DARO, AI, ILC)
Execution LDI				
0x01	5..8	X	X	Read(PCAO, AI, PCCE, ILC)
Execution STA				
0x02	5..8	X	X	Read(PCAO, DARI, PCCE)
0x02	9..12	X	X	Write(DARO, AO, ILC)
Execution LIA				
0x03	5	X	X	ACO, AO, DARI
0x03	6..9	X	X	Read(DARO, AI, ILC)
Execution NOT				
0x04	5	X	X	ACO, MODE, RO, AI, ILC
Execution AND				
0x05	5..8	X	X	Read(PCAO, DARI, PCCE)
0x05	9..12	X	X	Read(DARO, BI)
0x05	13	X	X	ACO, MODE, S3, S1, S0, RO, AI, ILC
Execution OR				
0x06	5..8	X	X	Read(PCAO, DARI, PCCE)
0x06	9..12	X	X	Read(DARO, BI)
0x06	13	X	X	ACO, MODE, S3, S2, S1, RO, AI, ILC
Execution XOR				
0x07	5..8	X	X	Read(PCAO, DARI, PCCE)
0x07	9..12	X	X	Read(DARO, BI)
0x07	13	X	X	ACO, MODE, S2, S1, RO, AI, ILC
Execution SHL				
0x08	5	X	X	ACO, S3, S2, RO, AI, ILC
Execution SHR				
0x09	5	X	X	ACO, SHR, AI, ILC
Execution ADD				
0x0A	5..8	X	X	Read(PCAO, DARI, PCCE)
0x0A	9..12	X	X	Read(DARO, BI)
0x0A	13	X	X	ACO, S3, S0, RO, AI, ILC
Execution SUB				
0x0B	5..8	X	X	Read(PCAO, DARI, PCCE)
0x0B	9..12	X	X	Read(DARO, BI)
0x0B	13	X	X	ACO, CIN, S2, S1, RO, AI, ILC
Execution CMP				
0x0C	5..8	X	X	Read(PCAO, DARI, PCCE)
0x0C	9..12	X	X	Read(DARO, BI)
0x0C	13	X	X	ACO, S2, S1, RO, ILC
Execution JMP				
0x0D	5..8	X	X	Read(PCAO, PCI, ILC)
Execution JPE				
0x0E	5..8	1	1	Read(PCAO, PCI, ILC)
0x0E	5	0	X	ACO, PCCE, ILC
0x0E	5	X	0	ACO, PCCE, ILC
Execution JPA				
0x0F	5..8	0	0	Read(PCAO, PCI, ILC)
0x0F	5	1	X	ACO, PCCE, ILC
0x0F	5	X	1	ACO, PCCE, ILC
Execution JPB				
0x10	5..8	0	1	Read(PCAO, PCI, ILC)
0x10	5	1	X	ACO, PCCE, ILC
0x10	5	0	0	ACO, PCCE, ILC
Execution CALL				
0x11	5	X	X	ACO, PCDO, DARI, PCCE
0x11	6..9	X	X	Write(SPO, PCDO, SPCE)
0x11	10..13	X	X	Read(DARO, PCI, ILC)
Execution RET				
0x12	5	X	X	ACO, SPU/SPD, SPCE
0x12	6..9	X	X	Read(SPO, PCI, ILC)
Execution LSP				
0x13	5	X	X	ACO, AO, SPI, ILC
Execution PUSH				
0x14	5..8	X	X	Write(SPO, AO, SPCE, ILC)
Execution POP				
0x15	5	X	X	ACO, SPU/SPD, SPCE
0x15	6..9	X	X	Read(SPO, AI, ILC)
Execution IN				
0x16	5..8	X	X	Read(PCAO, DARI, PCCE)
0x16	9..12	X	X	Read(M/IO, DARO, AI, ILC)
Execution OUT				
0x17	5..8	X	X	Read(PCAO, DARI, PCCE)
0x17	9..12	X	X	Write(M/IO, DARO, AO, ILC)
Execution NOP				
0x18	4	X	X	ACO, PCCE, ILC
Execution HALT				
0x19	1	X	X	ACO, ILC
0x19	4	X	X	ACO, ILC

Microcode

LDA		
IC	Inputs	Outputs
1	0x004, 0x005, 0x006, 0x007	0x3CB73A1F
2	0x008, 0x009, 0x00A, 0x00B	0x36377A1F
3	0x00C, 0x00D, 0x00E, 0x00F	0x3637721F
4	0x010, 0x011, 0x012, 0x013	0x3EB7FA1F
5	0x014, 0x015, 0x016, 0x017	0x3CB73A1F
6	0x018, 0x019, 0x01A, 0x01B	0x36377A1F
7	0x01C, 0x01D, 0x01E, 0x01F	0x36277A1F
8	0x020, 0x021, 0x022, 0x023	0x3EB7FA1F
9	0x024, 0x025, 0x026, 0x027	0x3C977A1F
10	0x028, 0x029, 0x02A, 0x02B	0x36377A1F
11	0x02C, 0x02D, 0x02E, 0x02F	0x36377A1E
12	0x030, 0x031, 0x032, 0x033	0x1EB77A1F

NOT		
IC	Inputs	Outputs
1	0x104, 0x105, 0x106, 0x107	0x3CB73A1F
2	0x108, 0x109, 0x10A, 0x10B	0x36377A1F
3	0x10C, 0x10D, 0x10E, 0x10F	0x3637721F
4	0x110, 0x111, 0x112, 0x113	0x3EB7FA1F
5	0x114, 0x115, 0x116, 0x117	0x1EB77E16

SHL		
IC	Inputs	Outputs
1	0x204, 0x205, 0x206, 0x207	0x3CB73A1F
2	0x208, 0x209, 0x20A, 0x20B	0x36377A1F
3	0x20C, 0x20D, 0x20E, 0x20F	0x3637721F
4	0x210, 0x211, 0x212, 0x213	0x3EB7FA1F
5	0x214, 0x215, 0x216, 0x217	0x1EB77B96

CMP		
IC	Inputs	Outputs
1	0x304, 0x305, 0x306, 0x307	0x3CB73A1F
2	0x308, 0x309, 0x30A, 0x30B	0x36377A1F
3	0x30C, 0x30D, 0x30E, 0x30F	0x3637721F
4	0x310, 0x311, 0x312, 0x313	0x3EB7FA1F
5	0x314, 0x315, 0x316, 0x317	0x3CB73A1F
6	0x318, 0x319, 0x31A, 0x31B	0x36377A1F
7	0x31C, 0x31D, 0x31E, 0x31F	0x36277A1F
8	0x320, 0x321, 0x322, 0x323	0x3EB7FA1F
9	0x324, 0x325, 0x326, 0x327	0x3C977A1F
10	0x328, 0x329, 0x32A, 0x32B	0x36377A1F
11	0x32C, 0x32D, 0x32E, 0x32F	0x36377A1B
12	0x330, 0x331, 0x332, 0x333	0x3EB77A1F
13	0x334, 0x335, 0x336, 0x337	0x1EB77AD7

JPB		
IC	Inputs	Outputs
1	0x404, 0x405, 0x406, 0x407	0x3CB73A1F
2	0x408, 0x409, 0x40A, 0x40B	0x36377A1F
3	0x40C, 0x40D, 0x40E, 0x40F	0x3637721F
4	0x410, 0x411, 0x412, 0x413	0x3EB7FA1F
5	0x415	0x3CB73A1F
5	0x416, 0x417, 0x414	0x1EB7FA1F
6	0x419	0x36377A1F
7	0x41D	0x36376A1F
8	0x421	0x1EB77A1F

PUSH		
IC	Inputs	Outputs
1	0x504, 0x505, 0x506, 0x507	0x3CB73A1F
2	0x508, 0x509, 0x50A, 0x50B	0x36377A1F
3	0x50C, 0x50D, 0x50E, 0x50F	0x3637721F
4	0x510, 0x511, 0x512, 0x513	0x3EB7FA1F
5	0x514, 0x515, 0x516, 0x517	0x3CF57A1F
6	0x518, 0x519, 0x51A, 0x51B	0x2E777A1D
7	0x51C, 0x51D, 0x51E, 0x51F	0x3E777A1D
8	0x520, 0x521, 0x522, 0x523	0x1EB37A1F

NOP		
IC	Inputs	Outputs
1	0x604, 0x605, 0x606, 0x607	0x3CB73A1F
2	0x608, 0x609, 0x60A, 0x60B	0x36377A1F
3	0x60C, 0x60D, 0x60E, 0x60F	0x3637721F
4	0x610, 0x611, 0x612, 0x613	0x1EB7FA1F

LDI		
IC	Inputs	Outputs
1	0x044, 0x045, 0x046, 0x047	0x3CB73A1F
2	0x048, 0x049, 0x04A, 0x04B	0x36377A1F
3	0x04C, 0x04D, 0x04E, 0x04F	0x3637721F
4	0x050, 0x051, 0x052, 0x053	0x3EB7FA1F
5	0x054, 0x055, 0x056, 0x057	0x3CB73A1F
6	0x058, 0x059, 0x05A, 0x05B	0x36377A1F
7	0x05C, 0x05D, 0x05E, 0x05F	0x36377A1E
8	0x060, 0x061, 0x062, 0x063	0x1EB7FA1F

AND		
IC	Inputs	Outputs
1	0x144, 0x145, 0x146, 0x147	0x3CB73A1F
2	0x148, 0x149, 0x14A, 0x14B	0x36377A1F
3	0x14C, 0x14D, 0x14E, 0x14F	0x3637721F
4	0x150, 0x151, 0x152, 0x153	0x3EB7FA1F
5	0x154, 0x155, 0x156, 0x157	0x3CB73A1F
6	0x158, 0x159, 0x15A, 0x15B	0x36377A1F
7	0x15C, 0x15D, 0x15E, 0x15F	0x36277A1F
8	0x160, 0x161, 0x162, 0x163	0x3EB7FA1F
9	0x164, 0x165, 0x166, 0x167	0x3C977A1F
10	0x168, 0x169, 0x16A, 0x16B	0x36377A1F
11	0x16C, 0x16D, 0x16E, 0x16F	0x36377A1B
12	0x170, 0x171, 0x172, 0x173	0x3EB77A1F
13	0x174, 0x175, 0x176, 0x177	0x1EB77F76

SHR		
IC	Inputs	Outputs
1	0x244, 0x245, 0x246, 0x247	0x3CB73A1F
2	0x248, 0x249, 0x24A, 0x24B	0x36377A1F
3	0x24C, 0x24D, 0x24E, 0x24F	0x3637721F
4	0x250, 0x251, 0x252, 0x253	0x3EB7FA1F
5	0x254, 0x255, 0x256, 0x257	0x1EB77A0E

JMP		
IC	Inputs	Outputs
1	0x344, 0x345, 0x346, 0x347	0x3CB73A1F
2	0x348, 0x349, 0x34A, 0x34B	0x36377A1F
3	0x34C, 0x34D, 0x34E, 0x34F	0x3637721F
4	0x350, 0x351, 0x352, 0x353	0x3EB7FA1F
5	0x354, 0x355, 0x356, 0x357	0x3CB73A1F
6	0x358, 0x359, 0x35A, 0x35B	0x36377A1F
7	0x35C, 0x35D, 0x35E, 0x35F	0x36376A1F
8	0x360, 0x361, 0x362, 0x363	0x1EB77A1F

CALL		
IC	Inputs	Outputs
1	0x444, 0x445, 0x446, 0x447	0x3CB73A1F
2	0x448, 0x449, 0x44A, 0x44B	0x36377A1F
3	0x44C, 0x44D, 0x44E, 0x44F	0x3637721F
4	0x450, 0x451, 0x452, 0x453	0x3EB7FA1F
5	0x454, 0x455, 0x456, 0x457	0x3EA7DA1F
6	0x458, 0x459, 0x45A, 0x45B	0x3CF57A1F
7	0x45C, 0x45D, 0x45E, 0x45F	0x2E775A1F
8	0x460, 0x461, 0x462, 0x463	0x3E775A1F
9	0x464, 0x465, 0x466, 0x467	0x3EB37A1F
10	0x468, 0x469, 0x46A, 0x46B	0x3C977A1F
11	0x46C, 0x46D, 0x46E, 0x46F	0x36377A1F
12	0x470, 0x471, 0x472, 0x473	0x36376A1F
13	0x474, 0x475, 0x476, 0x477	0x1EB77A1F

POP		
IC	Inputs	Outputs
1	0x544, 0x545, 0x546, 0x547	0x3CB73A1F
2	0x548, 0x549, 0x54A, 0x54B	0x36377A1F
3	0x54C, 0x54D, 0x54E, 0x54F	0x3637721F
4	0x550, 0x551, 0x552, 0x553	0x3EB7FA1F
5	0x554, 0x555, 0x556, 0x557	0x3EBB7A1F
6	0x558, 0x559, 0x55A, 0x55B	0x3CB57A1F
7	0x55C, 0x55D, 0x55E, 0x55F	0x36377A1F
8	0x560, 0x561, 0x562, 0x563	0x36377A1E
9	0x564, 0x565, 0x566, 0x567	0x1EB77A1F

HALT		
IC	Inputs	Outputs
1	0x644, 0x645, 0x646, 0x647	0x1EB77A1F
2	0x648, 0x649, 0x64A, 0x64B	0x36377A1F
3	0x64C, 0x64D, 0x64E, 0x64F	0x3637721F
4	0x650, 0x651, 0x652, 0x653	0x1EB77A1F

STA		
IC	Inputs	Outputs
1	0x084, 0x085, 0x086, 0x087	0x3CB73A1F
2	0x088, 0x089, 0x08A, 0x08B	0x36377A1F
3	0x08C, 0x08D, 0x08E, 0x08F	0x3637721F
4	0x090, 0x091, 0x092, 0x093	0x3EB7FA1F
5	0x094, 0x095, 0x096, 0x097	0x3CB73A1F
6	0x098, 0x099, 0x09A, 0x09B	0x36377A1F
7	0x09C, 0x09D, 0x09E, 0x09F	0x36277A1F
8	0x0A0, 0x0A1, 0x0A2, 0x0A3	0x3EB7FA1F
9	0x0A4, 0x0A5, 0x0A6, 0x0A7	0x3CD77A1F
10	0x0A8, 0x0A9, 0x0AA, 0x0AB	0x2E777A1D
11	0x0AC, 0x0AD, 0x0AE, 0x0AF	0x3E777A1D
12	0x0B0, 0x0B1, 0x0B2, 0x0B3	0x1EB77A1F

OR		
IC	Inputs	Outputs
1	0x184, 0x185, 0x186, 0x187	0x3CB73A1F
2	0x188, 0x189, 0x18A, 0x18B	0x36377A1F
3	0x18C, 0x18D, 0x18E, 0x18F	0x3637721F
4	0x190, 0x191, 0x192, 0x193	0x3EB7FA1F
5	0x194, 0x195, 0x196, 0x197	0x3CB73A1F
6	0x198, 0x199, 0x19A, 0x19B	0x36377A1F
7	0x19C, 0x19D, 0x19E, 0x19F	0x36277A1F
8	0x1A0, 0x1A1, 0x1A2, 0x1A3	0x3EB7FA1F
9	0x1A4, 0x1A5, 0x1A6, 0x1A7	0x3C977A1F
10	0x1A8, 0x1A9, 0x1AA, 0x1AB	0x36377A1F
11	0x1AC, 0x1AD, 0x1AE, 0x1AF	0x36377A1B
12	0x1B0, 0x1B1, 0x1B2, 0x1B3	0x3EB77A1F
13	0x1B4, 0x1B5, 0x1B6, 0x1B7	0x1EB77FD6

ADD		
IC	Inputs	Outputs
1	0x284, 0x285, 0x286, 0x287	0x3CB73A1F
2	0x288, 0x289, 0x28A, 0x28B	0x36377A1F
3	0x28C, 0x28D, 0x28E, 0x28F	0x3637721F
4	0x290, 0x291, 0x292, 0x293	0x3EB7FA1F
5	0x294, 0x295, 0x296, 0x297	0x3CB73A1F
6	0x298, 0x299, 0x29A, 0x29B	0x36377A1F
7	0x29C, 0x29D, 0x29E, 0x29F	0x36277A1F
8	0x2A0, 0x2A1, 0x2A2, 0x2A3	0x3EB7FA1F
9	0x2A4, 0x2A5, 0x2A6, 0x2A7	0x3C977A1F
10	0x2A8, 0x2A9, 0x2AA, 0x2AB	0x36377A1F
11	0x2AC, 0x2AD, 0x2AE, 0x2AF	0x36377A1B
12	0x2B0, 0x2B1, 0x2B2, 0x2B3	0x3EB77A1F
13	0x2B4, 0x2B5, 0x2B6, 0x2B7	0x1EB77B36

JPE		
IC	Inputs	Outputs
1	0x384, 0x385, 0x386, 0x387	0x3CB73A1F
2	0x388, 0x389, 0x38A, 0x38B	0x36377A1F
3	0x38C, 0x38D, 0x38E, 0x38F	0x3637721F
4	0x390, 0x391, 0x392, 0x393	0x3EB7FA1F
5	0x397	0x3CB73A1F
5	0x394, 0x395, 0x394, 0x396	0x1EB7FA1F
6	0x39B	0x36377A1F
7	0x39F	0x36376A1F
8	0x3A3	0x1EB77A1F

RET		
IC	Inputs	Outputs
1	0x484, 0x485, 0x486, 0x487	0x3CB73A1F
2	0x488, 0x489, 0x48A, 0x48B	0x36377A1F
3	0x48C, 0x48D, 0x48E, 0x48F	0x3637721F
4	0x490, 0x491, 0x492, 0x493	0x3EB7FA1F
5	0x494, 0x495, 0x496, 0x497	0x3EBB7A1F
6	0x498, 0x499, 0x49A, 0x49B	0x3CB57A1F
7	0x49C, 0x49D, 0x49E, 0x49F	0x36377A1F
8	0x4A0, 0x4A1, 0x4A2, 0x4A3	0x36376A1F
9	0x4A4, 0x4A5, 0x4A6, 0x4A7	0x1EB77A1F

IN		
IC	Inputs	Outputs
1	0x584, 0x585, 0x586, 0x587	0x3CB73A1F
2	0x588, 0x589, 0x58A, 0x58B	0x36377A1F
3	0x58C, 0x58D, 0x58E, 0x58F	0x3637721F
4	0x590, 0x591, 0x592, 0x593	0x3EB7FA1F
5	0x594, 0x595, 0x596, 0x597	0x3CB73A1F
6	0x598, 0x599, 0x59A, 0x59B	0x36377A1F
7	0x59C, 0x59D, 0x59E, 0x59F	0x36277A1F
8	0x5A0, 0x5A1, 0x5A2, 0x5A3	0x3EB7FA1F
9	0x5A4, 0x5A5, 0x5A6, 0x5A7	0x38977A1F
10	0x5A8, 0x5A9, 0x5AA, 0x5AB	0x32377A1F
11	0x5AC, 0x5AD, 0x5AE, 0x5AF	0x32377A1E
12	0x5B0, 0x5B1, 0x5B2, 0x5B3	0x1AB77A1F

OUT		
IC	Inputs	Outputs
1	0x5C4, 0x5C5, 0x5C6, 0x5C7	0x3CB73A1F
2	0x5C8, 0x5C9, 0x5CA, 0x5CB	0x36377A1F
3	0x5CC, 0x5CD, 0x5CE, 0x5CF	0x3637721F
4	0x5D0, 0x5D1, 0x5D2, 0x5D3	0x3EB7FA1F
5	0x5D4, 0x5D5, 0x5D6, 0x5D7	0x3CB73A1F
6	0x5D8, 0x5D9, 0x5DA, 0x5DB	0x36377A1F
7	0x5DC, 0x5DD, 0x5DE, 0x5DF	0x36277A1F
8	0x5E0, 0x5E1, 0x5E2, 0x5E3	0x3EB7FA1F
9	0x5E4, 0x5E5, 0x5E6, 0x5	